Diag Solutions

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1. **1.** For the equation $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$, if we substitute $u = \frac{1}{x}$ and $v = \frac{1}{y}$, what will be the resulting linear equation from the first expression?

Answer: (A)

Solution:

1. Substitute $\frac{1}{x} = u$ and $\frac{1}{y} = v$ into the given equation.
2. The term $\frac{2}{x}$ becomes $2u$.
3. The term $\frac{3}{y}$ becomes $3v$.
4. The resulting equation is $2u + 3v = 13$.

Derivation:

$$2(\frac{1}{x}) + 3(\frac{1}{y}) = 13$$
$$\implies 2u + 3v = 13$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

2. **2.** If $\frac{1}{x+y} = a$ and $\frac{1}{x-y} = b$, the equation $\frac{5}{x+y} - \frac{2}{x-y} = 1$ reduces to which of the following linear forms?

Answer: (C)

Solution:

1. Substitute a for $\frac{1}{x+y}$ and b for $\frac{1}{x-y}$.
2. Substitute the variables into the equation $5(\frac{1}{x+y}) - 2(\frac{1}{x-y}) = 1$.
3. The resulting equation is $5a - 2b = 1$.

Derivation:

$$5a - 2b = 1$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

3. **3.** Given the equation $\frac{7}{x-1} + \frac{1}{y-2} = 3$, what substitution would convert this into a linear equation?

Answer: (B)

Solution:

1. Identify the variable parts in the denominators.
2. Let $u = \frac{1}{x-1}$ and $v = \frac{1}{y-2}$.
3. The equation becomes $7u + v = 3$, which is linear in u and v.

Derivation:

$$u = \frac{1}{x-1}, v = \frac{1}{y-2}$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

4. **4.** For the system $\frac{10}{x+y} + \frac{2}{x-y} = 4$ and $\frac{15}{x+y} - \frac{5}{x-y} = -2$, if $u = \frac{1}{x+y}$ and $v = \frac{1}{x-y}$, what is the coefficient of v in the second equation?

   Answer: (D)

   Solution:

   1. Substitute $v = \frac{1}{x-y}$ into the second equation $\frac{15}{x+y} - \frac{5}{x-y} = -2$.
   2. This results in $15u - 5v = -2$.
   3. The coefficient of v is -5.

   Derivation:

$$\begin{aligned} 15u - 5v &= -2 \\ \implies \text{coefficient of } v &= -5 \end{aligned}$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

5. **5.** Which of the following represents the linear form of $\frac{x+y}{xy} = 2$?

   Answer: (A)

   Solution:

   1. Divide each term in the numerator by the denominator: $\frac{x}{xy} + \frac{y}{xy} = 2$.
   2. Simplify the fractions: $\frac{1}{y} + \frac{1}{x} = 2$.
   3. This is equivalent to $\frac{1}{x} + \frac{1}{y} = 2$.

   Derivation:

$$\begin{aligned} \frac{x}{xy} + \frac{y}{xy} &= 2 \\ \implies \frac{1}{y} + \frac{1}{x} &= 2 \end{aligned}$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

6. **6.** Assertion (A): The equation $\frac{2}{x} + \frac{3}{y} = 13$ can be reduced to a linear equation in two variables by substituting $u = \frac{1}{x}$ and $v = \frac{1}{y}$. Reason (R): Any equation of the form $\frac{a}{x} + \frac{b}{y} = c$ where $x, y \neq 0$ is a linear equation. Select the correct option.

   Answer: (C)

   Solution:

   1. The substitution $u = 1/x$ and $v = 1/y$ converts $2/x + 3/y = 13$ into $2u + 3v = 13$, which is linear. Thus (A) is true.
   2. The equation $a/x + b/y = c$ is not linear as the variables are in the denominator. Thus (R) is false.

Derivation:

$$
\begin{aligned}
2u + 3v &= 13 \\
\implies &\text{ Linear form}
\end{aligned}
$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

7. **7.** Assertion (A): For the system $\frac{1}{x}+\frac{1}{y}=5$ and $\frac{1}{x}-\frac{1}{y}=1$, the values are $x=1/3, y=1/2$. Reason (R): Adding the two equations eliminates the term involving $1/y$. Select the correct option.

   Answer: (A)

   Solution:

   1. Let $u=1/x, v=1/y$. Equations become $u+v=5$ and $u-v=1$.
   2. Adding gives $2u=6 \implies u=3 \implies x=1/3$.
   3. Subtracting gives $2v=4 \implies v=2 \implies y=1/2$. (A) is true.
   4. Adding the equations eliminates v. (R) is true and explains (A).

   Derivation:

$$
\begin{aligned}
u+v=5, u-v&=1 \\
\implies 2u &= 6 \\
\implies u=3; 2v &= 4 \\
\implies v &= 2
\end{aligned}
$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

8. **8.** Assertion (A): The equation $\frac{5}{x-1}+\frac{1}{y-2}=2$ is reducible to linear form using $u=\frac{1}{x-1}$ and $v=\frac{1}{y-2}$. Reason (R): The substitution must always be linear functions of x and y. Select the correct option.

   Answer: (B)

   Solution:

   1. (A) is true because $5u+v=2$ is linear.
   2. (R) is true because we reduce to linear form to solve, but it is not the reason for the specific substitution choice in (A).

   Derivation:

$$
5(\frac{1}{x-1})+1(\frac{1}{y-2})=2
$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

9. **9.** Assertion (A): The pair $\frac{4}{x}+3y=14$ and $\frac{3}{x}-4y=23$ has $x=1/5$. Reason (R): Substituting $u=1/x$ reduces the system to $4u+3y=14$ and $3u-4y=23$. Select the correct option.

   Answer: (A)

   Solution:

1. (R) is true as it correctly defines the substitution.
2. Solving $4u + 3y = 14$ and $3u - 4y = 23$: Multiply first by 4, second by 3: $16u + 12y = 56$ and $9u - 12y = 69$.
3. Adding: $25u = 125 \implies u = 5 \implies 1/x = 5 \implies x = 1/5$. (A) is true.

Derivation:

$$\begin{aligned} 16u + 12y = 56; 9u - 12y = 69 \\ \implies 25u = 125 \\ \implies u = 5 \end{aligned}$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

10. **10.** Assertion (A): For $\frac{2}{\sqrt{x}} + \frac{3}{\sqrt{y}} = 2$ and $\frac{4}{\sqrt{x}} - \frac{9}{\sqrt{y}} = -1$, the value of $u = \frac{1}{\sqrt{x}}$ is 1/2. Reason (R): A linear equation must have variables with power 1. Select the correct option.

Answer: (B)

Solution:

1. (A) is true: $2u + 3v = 2$ and $4u - 9v = -1$. Multiply 1st by 3: $6u + 9v = 6$. Adding to 2nd: $10u = 5 \implies u = 1/2$.
2. (R) is true as linear equations are defined by degree 1, but it is not the reason why $u = 1/2$.

Derivation:

$$\begin{aligned} 2u + 3v = 2; 4u - 9v = -1 \\ \implies 6u + 9v = 6 \\ \implies 10u = 5 \\ \implies u = 1/2 \end{aligned}$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

11. **11.** Solve for x and y: $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$.

Answer:

$$x = \frac{1}{2}, y = \frac{1}{3}$$

Solution:

1. Substitute $u = 1/x$ and $v = 1/y$ to convert the equations into standard linear form: $2u + 3v = 13$ and $5u - 4v = -2$.
2. Solve the system using elimination method to find $u = 2$ and $v = 3$, then invert to find x and y.

Derivation: Multiply eq1 by 4 and eq2 by 3: 8u + 12v = 52 and 15u - 12v = -6 . Adding them gives 23u = 46 $\implies u = 2$. Thus, 1/x = 2 $\implies x = 1/2$. Substituting u=2 into 2(2) + 3v = 13 $\implies 3v = 9 \implies v = 3$. Thus, 1/y = 3 $\implies y = 1/3$. *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

12. **12.** Find the values of x and y given: $\frac{1}{2x} - \frac{1}{y} = -1$ and $\frac{1}{x} + \frac{1}{2y} = 8$.

Answer:

$$x = 1/6, y = 1/14$$

Solution:

1. Let $u = 1/x$ and $v = 1/y$. The equations become $\frac{1}{2}u - v = -1$ and $u + \frac{1}{2}v = 8$.
2. Solve for u and v and find their reciprocals.

Derivation: Multiply eq1 by 2: u - 2v = -2 . Multiply eq2 by 2: 2u + v = 16 . From eq2, v = 16 - 2u . Substitute into eq1: u - 2(16 - 2u) = -2 $\implies u - 32 + 4u = -2 \implies 5u = 30 \implies u = 6$. Then v = 16 - 12 = 4 . So x = 1/6, y = 1/4 . *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

13. **13.** Solve the system: $\frac{10}{x+y} + \frac{2}{x-y} = 4$ and $\frac{15}{x+y} - \frac{5}{x-y} = -2$.

Answer:

$$x = 3, y = 2$$

Solution:

1. Let $u = 1/(x + y)$ and $v = 1/(x - y)$. The equations become $10u + 2v = 4$ and $15u - 5v = -2$.
2. Solve for u, v and equate to $(x + y)$ and $(x - y)$ to solve for x and y.

Derivation: Multiply first by 5 and second by 2: 50u + 10v = 20 and 30u - 10v = -4 . Add: 80u = 16 $\implies u = 1/5$. Thus x+y=5 . Substitute u=1/5 in 10(1/5) + 2v = 4 $\implies 2 + 2v = 4 \implies v = 1$. Thus x-y=1 . Adding gives 2x=6, x=3 ; subtracting gives 2y=4, y=2 . *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

14. **14.** Solve for x and y: $\frac{4}{x} + 3y = 8$ and $\frac{6}{x} - 4y = -5$.

Answer:

$$x = 2, y = 0$$

Solution:

1. Let $u = 1/x$. The equations become $4u + 3y = 8$ and $6u - 4y = -5$.
2. Eliminate y by multiplying the first by 4 and the second by 3.

Derivation: Eq1: 16u + 12y = 32 . Eq2: 18u - 12y = -15 . Adding: 34u = 17 $\implies u = 1/2$. Since 1/x = 1/2, x = 2 . Substitute u=1/2 into 4(1/2) + 3y = 8 $\implies 2 + 3y = 8 \implies 3y = 6 \implies y = 2$. *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

15. **15.** Solve: $\frac{1}{3x+y} + \frac{1}{3x-y} = \frac{3}{4}$ and $\frac{1}{2(3x+y)} - \frac{1}{2(3x-y)} = -\frac{1}{8}$.

Answer:

$$x = 1, y = 1$$

Solution:

1. Let $u = 1/(3x+y)$ and $v = 1/(3x-y)$. Equations are $u + v = 3/4$ and $u/2 - v/2 = -1/8$.
2. Solve for u and v and then solve the resulting system for x and y.

Derivation: Eq1: u + v = 3/4 . Eq2: u - v = -1/4 . Adding gives 2u = 2/4 = 1/2 $\implies u = 1/4$. Then v = 1/2 . Thus 3x+y=4 and 3x-y=2 . Adding gives 6x=6 $\implies x = 1$. Substituting gives 3(1)+y=4 $\implies y = 1$. *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

16. **16.** Solve the following pair of equations by reducing them to a linear form: $\frac{2}{x}+\frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$.

Answer:

$$x = \frac{1}{2}, y = \frac{1}{3}$$

Solution:

1. Substitute $u = \frac{1}{x}$ and $v = \frac{1}{y}$ to transform the equations into linear form.
2. The equations become $2u + 3v = 13$ and $5u - 4v = -2$.
3. Solve the system of linear equations using the elimination method.
4. Back-substitute the values of u and v to find x and y.

Derivation: Multiply first by 4 and second by 3: 8u + 12v = 52 and 15u - 12v = -6 . Adding them gives 23u = 46 , so u = 2 . Then 2(2) + 3v = 13 $\implies 3v = 9 \implies v = 3$. Thus x = 1/2, y = 1/3 . *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

17. **17.** A boat goes 30 km upstream and 44 km downstream in 10 hours. In 13 hours, it can go 40 km upstream and 55 km downstream. Determine the speed of the stream and the speed of the boat in still water.

Answer:

$$\text{Speed of boat} = 8km/hr, \text{Speed of stream} = 3km/hr$$

Solution:

1. Let speed of boat be x km/hr and stream be y km/hr. Upstream speed is $x - y$, downstream is $x + y$.
2. Form equations: $\frac{30}{x-y} + \frac{44}{x+y} = 10$ and $\frac{40}{x-y} + \frac{55}{x+y} = 13$.
3. Substitute $u = \frac{1}{x-y}$ and $v = \frac{1}{x+y}$ and solve for u and v.
4. Calculate x and y from u and v.

Derivation: 30u + 44v = 10 $\implies$ $15u + 22v = 5$ and 40u + 55v = 13 $\implies$ $8u + 11v = 2.6$. Solving gives u=1/5, v=1/11 . Thus x-y=5, x+y=11 . Adding gives 2x=16 $\implies$ $x = 8, y = 3$. *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

18. **18.** Solve for x and y: $\frac{1}{2(2x+3y)} + \frac{12}{7(3x-2y)} = \frac{1}{2}$ and $\frac{7}{2x+3y} + \frac{4}{3x-2y} = 2$.

Answer:

$$x = 2, y = 1$$

Solution:

1. Let $u = \frac{1}{2x+3y}$ and $v = \frac{1}{3x-2y}$.
2. The equations become $\frac{1}{2}u + \frac{12}{7}v = \frac{1}{2}$ and $7u + 4v = 2$.
3. Simplify the first into $7u + 24v = 7$.
4. Solve the system $7u + 24v = 7$ and $7u + 4v = 2$ to find u and v, then solve for x, y.

Derivation: Subtracting equations: (24-4)v = 7-2 $\implies$ $20v = 5$ $\implies$ $v = 1/4$. Then 7u + 1 = 2 $\implies$ $u = 1/7$. 2x+3y=7, 3x-2y=4 . Solving gives x=2, y=1 . *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

19. **19.** Roohi travels 300 km to her home partly by train and partly by bus. She takes 4 hours if she travels 60 km by train and the rest by bus. If she travels 100 km by train and the rest by bus, she takes 10 minutes longer. Find the speed of the train and the bus.

Answer:

$$\text{Speed of train} = 60km/hr, \text{Speed of bus} = 80km/hr$$

Solution:

1. Let speed of train be x and bus be y. Time taken = distance/speed.
2. Equations: $\frac{60}{x} + \frac{240}{y} = 4$ and $\frac{100}{x} + \frac{200}{y} = 4 + \frac{1}{6}$.
3. Use substitution $u = 1/x, v = 1/y$ to solve the linear system.
4. Solve for x and y.

Derivation: 60u + 240v = 4 $\implies$ $15u + 60v = 1$. 100u + 200v = 25/6 $\implies$ $24u + 48v = 1$. Solving yields u=1/60, v=1/80 . Hence x=60, y=80 . *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

20. **20.** Solve the system: $\frac{x+y}{xy} = 2$ and $\frac{x-y}{xy} = 6$.

Answer:

$$x = -1/2, y = 1/4$$

Solution:

1. Divide by xy to get $\frac{1}{y} + \frac{1}{x} = 2$ and $\frac{1}{y} - \frac{1}{x} = 6$.
2. Let $u = \frac{1}{x}$ and $v = \frac{1}{y}$.
3. The system becomes $v + u = 2$ and $v - u = 6$.
4. Solve for u and v and then for x and y.

Derivation: Adding: 2v = 8 $\implies$ $v = 4$. Subtracting: 2u = -4 $\implies$ $u = -2$. Thus 1/y = 4 $\implies$ $y = 1/4$ and 1/x = -2 $\implies$ $x = -1/2$. *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

21. 21. Solve the following system of equations by reducing them to linear form: $\frac{10}{x+y} + \frac{2}{x-y} = 4$ and $\frac{15}{x+y} - \frac{5}{x-y} = -2$.

Answer:

$$x = 3, y = 2$$

Solution:

1. Let $u = \frac{1}{x+y}$ and $v = \frac{1}{x-y}$.
2. Substitute into equations to get: $10u + 2v = 4$ and $15u - 5v = -2$.
3. Multiply first eq by 5 and second by 2: $50u + 10v = 20$ and $30u - 10v = -4$.
4. Add equations to get $80u = 16$, so $u = 1/5$.
5. Substitute $u = 1/5$ into $10(1/5) + 2v = 4$ to get $2 + 2v = 4$, so $v = 1$.
6. Solve $x + y = 5$ and $x - y = 1$ to get $x = 3, y = 2$.

Derivation:

$$\begin{aligned}
10u + 2v &= 4 \\
\implies 5u + v &= 2 \\
\implies v &= 2 - 5u. \\
15u - 5(2 - 5u) &= -2 \\
\implies 15u - 10 + 25u &= -2 \\
\implies 40u &= 8 \\
\implies u &= 1/5. \\
v = 2 - 5(1/5) &= 1. \\
\frac{1}{x+y} &= 1/5 \\
\implies x + y = 5; \quad \frac{1}{x-y} &= 1 \\
\implies x - y &= 1. \\
(x + y) + (x - y) &= 5 + 1 \\
\implies 2x &= 6 \\
\implies x &= 3. \\
3 + y &= 5 \\
\implies y &= 2.
\end{aligned}$$

22. 22. A boat goes 30 km upstream and 44 km downstream in 10 hours. In 13 hours, it can go 40 km upstream and 55 km downstream. Determine the speed of the stream and that of the boat in still water.

Answer:

$$\text{Speed of boat} = 8km/hr, \text{Speed of stream} = 3km/hr$$

Solution:

1. Let speed of boat be x km/hr and stream be y km/hr.
2. Speed downstream is $x + y$, upstream is $x - y$.
3. Equations: $\frac{30}{x-y} + \frac{44}{x+y} = 10$ and $\frac{40}{x-y} + \frac{55}{x+y} = 13$.
4. Let $u = 1/(x - y)$ and $v = 1/(x + y)$.
5. Solve $30u + 44v = 10$ and $40u + 55v = 13$.
6. Find $u = 1/5, v = 1/11$, then solve $x - y = 5$ and $x + y = 11$.

Derivation: 30u + 44v = 10 $\implies 15u + 22v = 5$.
$40u + 55v = 13$.
Multiply (1) by 8: 120u + 176v = 40. \\ Multiply (2) by 3: 120u + 165v = 39. \\
Subtract: 11v = 1 $\implies v = 1/11$.
$15u + 2 = 5 \implies 15u = 3 \implies u = 1/5$.
$x + y = 11, x - y = 5 \implies 2x = 16, x = 8$; y=3. *Pair of Linear Equations in Two Variables — Equations reducible to linear form*

23. 23. Solve for x and y: $\frac{1}{2(2x+3y)} + \frac{12}{7(3x-2y)} = \frac{1}{2}$ and $\frac{7}{2x+3y} + \frac{4}{3x-2y} = 2$.

Answer:

$$x = 2, y = 1/3 (\text{or similar system solution})$$

Solution:

1. Let $u = \frac{1}{2x+3y}$ and $v = \frac{1}{3x-2y}$.
2. Substitute to get: $\frac{1}{2}u + \frac{12}{7}v = \frac{1}{2}$ and $7u + 4v = 2$.
3. Simplify first eq: $7u + 24v = 7$.
4. Subtract $(7u + 4v = 2)$ from $(7u + 24v = 7)$ to get $20v = 5 \implies v = 1/4$.
5. Substitute $v = 1/4$ into $7u + 4(1/4) = 2 \implies 7u = 1 \implies u = 1/7$.
6. Solve $2x + 3y = 7$ and $3x - 2y = 4$ using elimination.

Derivation:

$$\begin{aligned}2x + 3y &= 7 \ldots (i)\\ 3x - 2y &= 4 \ldots (ii)\\ (i) \times 2 &+ (ii) \times 3\\ \implies 4x + 6y + 9x - 6y &= 14 + 12\\ \implies 13x &= 26\\ \implies x &= 2.\\ 2(2) + 3y &= 7\\ \implies 3y &= 3\\ \implies y &= 1.\end{aligned}$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

24. **24.** Two women and five men can together finish an embroidery work in 4 days, while three women and six men can finish it in 3 days. Find the time taken by 1 woman alone to finish the work, and also that taken by 1 man alone.

Answer:

$$\text{Woman} = 18\text{days}, Man = 36\text{days}$$

Solution:

1. Let 1 woman take x days and 1 man take y days.
2. Work done by 1 woman in 1 day $= 1/x$, man $= 1/y$.
3. Equations: $4(2/x + 5/y) = 1$ and $3(3/x + 6/y) = 1$.
4. $8/x + 20/y = 1$ and $9/x + 18/y = 1$.
5. Let $u = 1/x, v = 1/y$. Solve $8u + 20v = 1$ and $9u + 18v = 1$.
6. Solve for u, v and reciprocate to find x, y.

Derivation:

$$\begin{aligned}8u + 20v &= 1 \ldots (1)\\ 9u + 18v &= 1 \ldots (2)\\ 72u + 180v &= 9\\ 72u + 144v &= 8\\ 36v &= 1\\ \implies v &= 1/36.\\ 9u + 18(1/36) &= 1\\ \implies 9u &= 1/2\\ \implies u &= 1/18.\\ x = 18, y &= 36.\end{aligned}$$

Pair of Linear Equations in Two Variables — Equations reducible to linear form

25. **25.** Which of the following is a quadratic equation?

Answer: (C)

Solution:

1. A quadratic equation is of the form $ax^2 + bx + c = 0$ where $a \neq 0$.
2. Check each option: (A) has x^2 on both sides cancelling out, (B) x^3 term makes it cubic, (C) simplifies to $x^2 - 3x + 2 = 0$, (D) has $1/x$ which is not polynomial.

Derivation:

$$\begin{aligned} x(x+1) + 8 &= (x+2)(x-2) \\ \implies x^2 + x + 8 &= x^2 - 4 \\ \implies x + 12 = 0 \text{ (Linear)}. \ (x+2)^2 &= 2(x+3) \\ \implies x^2 + 4x + 4 &= 2x + 6 \\ \implies x^2 + 2x - 2 &= 0 \end{aligned}$$

Quadratic Equations — Introduction to Quadratic Equations

26. **26.** The discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ is:

Answer: (B)

Solution:

1. Identify $a = 2, b = -4, c = 3$.
2. Use the formula $D = b^2 - 4ac$.
3. Calculate $D = (-4)^2 - 4(2)(3) = 16 - 24 = -8$.

Derivation:

$$D = b^2 - 4ac = (-4)^2 - 4(2)(3) = 16 - 24 = -8$$

Quadratic Equations — Nature of Roots

27. **27.** If $1/2$ is a root of the equation $x^2 - kx - 5/4 = 0$, then the value of k is:

Answer: (D)

Solution:

1. Substitute $x = 1/2$ into the equation.
2. $(1/2)^2 - k(1/2) - 5/4 = 0$.
3. $1/4 - k/2 - 5/4 = 0 \implies -4/4 - k/2 = 0 \implies -1 = k/2 \implies k = -2$.

Derivation:

$$\begin{aligned} \left(\frac{1}{2}\right)^2 - k\left(\frac{1}{2}\right) - \frac{5}{4} &= 0 \\ \implies \frac{1}{4} - \frac{k}{2} - \frac{5}{4} &= 0 \\ \implies -1 - \frac{k}{2} &= 0 \\ \implies k &= -2 \end{aligned}$$

Quadratic Equations — Roots of a Quadratic Equation

28. **28.** The quadratic equation $x^2 + kx + 1 = 0$ has equal roots if:

Answer: (A)

Solution:

1. For equal roots, discriminant $D = 0$.
2. $b^2 - 4ac = 0$.
3. $k^2 - 4(1)(1) = 0 \implies k^2 = 4 \implies k = \pm 2$.

Derivation:

$$\begin{aligned} D = k^2 - 4(1)(1) &= 0 \\ \implies k^2 &= 4 \\ \implies k &= \pm 2 \end{aligned}$$

Quadratic Equations — Nature of Roots

29. **29.** The roots of the quadratic equation $x^2 - 9 = 0$ are:

Answer: (C)

Solution:

1. Factor the equation: $x^2 - 3^2 = 0$.
2. $(x - 3)(x + 3) = 0$.
3. Therefore, $x = 3$ or $x = -3$.

Derivation:

$$\begin{aligned} x^2 &= 9 \\ \implies x &= \pm\sqrt{9} \\ \implies x &= \pm 3 \end{aligned}$$

Quadratic Equations — Solving Quadratic Equations by Factorization

30. **30.** Assertion (A): The quadratic equation $x^2 + 5x + 6 = 0$ has two distinct real roots. Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has distinct real roots if its discriminant $D = b^2 - 4ac > 0$.

Answer: (A)

Solution:

1. Calculate the discriminant D.
2. Check if D ¿ 0.

Derivation: D = bˆ2 - 4ac = (5)ˆ2 - 4(1)(6) = 25 - 24 = 1. Since 1 ¿ 0, the roots are distinct and real. *Quadratic Equations — Nature of Roots*

31. **31.** Assertion (A): If $x = 2$ is a root of $x^2 - kx + 4 = 0$, then $k = 4$. Reason (R): A value $x = \alpha$ is a root of the equation $f(x) = 0$ if $f(\alpha) = 0$.

Answer: (A)

Solution:

1. Substitute x = 2 into the equation.
2. Solve for k.

Derivation:

$$
\begin{aligned}
(2)^2 - k(2) + 4 &= 0 \\
\implies 4 - 2k + 4 &= 0 \\
\implies 8 &= 2k \\
\implies k &= 4.
\end{aligned}
$$

Quadratic Equations — Roots of Quadratic Equation

32. **32.** Assertion (A): The equation $(x-2)^2+1 = 2x-3$ is a quadratic equation. Reason (R): Any equation of the form $ax^2 + bx + c = 0$ where $a \neq 0$ is a quadratic equation.

Answer: (A)

Solution:

1. Expand the LHS.
2. Simplify the equation to standard form.
3. Check if the highest power is 2.

Derivation: xˆ2 - 4x + 4 + 1 = 2x - 3 $\implies$ $x^2 - 6x + 8 = 0$. This is in the form axˆ2 + bx + c = 0 with a = 1. *Quadratic Equations — Introduction to Quadratic Equations*

33. **33.** Assertion (A): The quadratic equation $x^2 + 2x + 5 = 0$ has no real roots. Reason (R): For $x^2 + 2x + 5 = 0$, the discriminant is -16.

Answer: (A)

Solution:

1. Calculate discriminant D.
2. Evaluate nature of roots based on D.

Derivation: D = bˆ2 - 4ac = (2)ˆ2 - 4(1)(5) = 4 - 20 = -16. Since D ¡ 0, there are no real roots. *Quadratic Equations — Nature of Roots*

34. **34.** Assertion (A): The roots of $x^2 - 9 = 0$ are 3 and -3. Reason (R): $x^2 - a^2 = (x-a)(x+a)$.

Answer: (A)

Solution:

1. Use factorisation method.
2. Solve for x.

Derivation:

$$
\begin{aligned}
x^2 - 9 &= 0 \\
\implies (x-3)(x+3) &= 0. So, x = 3 \text{or x} = -3.
\end{aligned}
$$

35. **35.** Find the roots of the quadratic equation $x^2 - 3x - 10 = 0$ by the method of factorization.

Answer:

$$x = 5, x = -2$$

Solution:

1. Split the middle term $-3x$ into $-5x + 2x$ such that their product is $-10x^2$.
2. Factorize the expression by grouping terms and solve for x.

Derivation:

$$\begin{aligned} x^2 - 5x + 2x - 10 = 0 \\ \implies x(x-5) + 2(x-5) = 0 \\ \implies (x-5)(x+2) = 0 \\ \implies x = 5, -2 \end{aligned}$$

Quadratic Equations — Factorization Method

36. **36.** Find the value of k for which the quadratic equation $2x^2 + kx + 3 = 0$ has two equal real roots.

Answer:

$$k = \pm 2\sqrt{6}$$

Solution:

1. Use the condition for equal roots, which is the discriminant $D = b^2 - 4ac = 0$.
2. Substitute $a = 2, b = k, c = 3$ into the formula and solve for k.

Derivation:

$$\begin{aligned} D = k^2 - 4(2)(3) = 0 \\ \implies k^2 - 24 = 0 \\ \implies k^2 = 24 \\ \implies k = \pm\sqrt{24} = \pm 2\sqrt{6} \end{aligned}$$

Quadratic Equations — Nature of Roots

37. **37.** Check whether $x(x+3) + 1 = (x-2)(x+2)$ is a quadratic equation.

Answer:

$$No, \text{it is not a quadratic equation.}$$

Solution:

1. Expand both sides of the equation to simplify.
2. Check if the resulting equation is of the form $ax^2 + bx + c = 0$ where $a \neq 0$.

Derivation:

$$x^2 + 3x + 1 = x^2 - 4$$
$$\implies 3x + 1 = -4$$
$$\implies 3x + 5 = 0. \text{ Since the degree is 1, it is linear, not quadratic.}$$

Quadratic Equations — Introduction to Quadratic Equations

38. **38.** Find the roots of the equation $2x^2 - x + \frac{1}{8} = 0$ by using the quadratic formula.

Answer:

$$x = \frac{1}{4}$$

Solution:

1. Multiply the entire equation by 8 to simplify the coefficients.
2. Apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Derivation:

$$16x^2 - 8x + 1 = 0. \text{ Here } a = 16, b = -8, c = 1.$$
$$\implies x = \frac{8 \pm \sqrt{64 - 64}}{32} = \frac{8}{32} = \frac{1}{4}$$

Quadratic Equations — Quadratic Formula

39. **39.** The sum of two numbers is 27 and their product is 182. Form the quadratic equation representing this scenario.

Answer:

$$x^2 - 27x + 182 = 0$$

Solution:

1. Let one number be x, then the other number is $27 - x$.
2. Set up the equation based on their product $x(27 - x) = 182$ and simplify.

Derivation:

$$x(27 - x) = 182$$
$$\implies 27x - x^2 = 182$$
$$\implies x^2 - 27x + 182 = 0$$

Quadratic Equations — Word Problems

40. **40.** Find the roots of the quadratic equation $2x^2 - x + \frac{1}{8} = 0$ by the method of factorization.

Answer:

$$x = \frac{1}{4}, \frac{1}{4}$$

Solution:

1. Multiply the entire equation by 8 to eliminate the fraction: $16x^2 - 8x + 1 = 0$.
2. Split the middle term: $16x^2 - 4x - 4x + 1 = 0$.
3. Factor by grouping: $4x(4x - 1) - 1(4x - 1) = 0$.
4. Solve for x: $(4x - 1)^2 = 0$, thus $x = 1/4$.

Derivation:

$$\begin{aligned} 16x^2 - 8x + 1 &= 0 \\ 16x^2 - 4x - 4x + 1 &= 0 \\ 4x(4x - 1) - 1(4x - 1) &= 0 \\ (4x - 1)(4x - 1) &= 0 \\ x &= 1/4 \end{aligned}$$

Quadratic Equations — Factorization Method